ON EQUIDISTRIBUTED BINARY NECKLACES

ADAM QUINN JAFFE

Abstract. We define a set of combinatorial objects known as binary necklaces and identify a subset that satisfy a certain equidistribution property. We prove that such an equidistribution is uniquely achieved in a configuration related to Christoffel words, and then provide a fast algorithm (linear in time and space) for checking whether a given binary necklace is equidistributed. Our main tool is the construction of a lossless compression map that preserves equidistribution.

1. Introduction

Notions of equidistribution in the circle have been central to many important questions across mathematical disciplines. In the context of number theory, Weyl [9] considered equidistribution of sequences modulo 1, and Erdős, and Hall [3] considered the equidistribution of arguments of the Gaussian integers. In combinatorics and geometry, Erdős and Turán [4] developed the field of discrepancy theory, which later became instrumental in the methods of quasi-Monte-Carlo integration.

Today, such questions have attracted attention, not only from the mathematics community, but also from scholars in computational geometry and computational music theory ([5],[7],[8]). In this domain, binary necklaces are a natural object of study, as they encode the placement of rhythmic or melodic onsets into discretized positions in a phrase. Empirical results (say, of [8]) show that equidistribution is, in fact, a quite favorable property in constructing rhythms that are interesting to the human ear.

However, there has been very little consideration given to the mathematical or algorithmic theory underlying these notions of equidistribution. The contents of this paper aim, not only to prove results of interests to the mathematical community, but also to bridge this gap. That being said, one can appreciate the contents of this paper with no reference to its applications to computational music theory.

We now give an overview of the notation and conventions used in this paper.

Conventions for the circle. Consider indices $m, n \in \mathbb{Z}/k\mathbb{Z}$, and let m_0 and n_0 denote the unique representatives of these equivalence classes that lie in the real interval $[0, k)$. We use $[m, n)_{\mathbb{S}^1}$ to denote the cyclic interval between m and n: If $m_0 \leq n_0$, this is exactly the real interval $[m, n)$; if $m_0 > n_0$, this notation is understood to mean $[n_0, m_0 + k)$. We use $d(m, n)$ to denote the length of $[m, n)_{\mathbb{S}^1}$, which in particular has $d(m, n) + d(n, m) = k$. Usually the subscript $\mathbb{S}^1$ will be clear from context, and we will omit it for brevity.

2010 *Mathematics Subject Classification.* Primary 05B25, 68R15; Secondary 05A05.

Key words and phrases. equidistribution, discrepancy theory, binary necklaces, computational music theory.

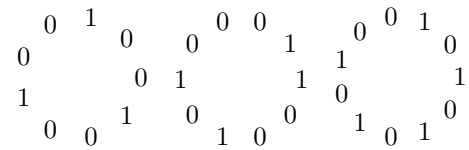

FIGURE 1. Examples of binary necklaces, with types (6,3) (left), (6,4) (center), and (6,5) (right).

Conventions for strings. For a string $\tilde{S}$ of length k, we use $\tilde{S}[r]$ to represent the element at index r. We use $\tilde{S}[m,n)$ to denote the substring corresponding to the indices in the half-open interval $[m,n)$, and likewise for other types of intervals. When binary strings are written explicitly, we will use superscripts to denote blocks of the same character (e.g. $1^2 0^2 1 0^3 = 11001000$). We use the symbol $\oplus$ to denote the bitwise XOR of two binary strings.

Miscellaneous conventions. We will often use the identity $\frac{x \mod y}{y} = \frac{x}{y} - \lfloor \frac{x}{y} \rfloor$. We will also make use of the fact that $\lceil x \rceil - \lfloor x \rfloor = 0$ iff x is an integer and $\lceil x \rceil - \lfloor x \rfloor = 1$ iff x is not an integer. We use $\#A$ to denote the cardinality of a set A. We use ϕ to denote Euler's totient function, $\phi(n) = \{0 \leq m \leq n : \gcd(m,n) = 1\}$.

2. Theory

Consider the action of the cyclic group $\mathbb{Z}/k\mathbb{Z}$ on the set $BS(k)$ of all binary sequences of length k, where $r \cdot \tilde{X}$ for $r \in \mathbb{Z}/k\mathbb{Z}$ and $\tilde{X} \in BS(k)$ is defined to be $(r \cdot \tilde{X})[i] = \tilde{X}[i+r]$. That is, the action cyclically permutes the elements of $\tilde{X}$ by r places to the left. This action allows us to define our basic object of study.

Definition 2.1. A *binary necklace* S is an orbit in $BS(k)$ under $\mathbb{Z}/k\mathbb{Z}$, and a binary sequence $\tilde{S}$ in this orbit is called an *unfolding* of S. The *type* of a binary necklace is the pair $(\#\tilde{S}^{-1}(0), \#\tilde{S}^{-1}(1))$ where $\tilde{S}$ is any unfolding of S.

We denote the type and length of a binary necklace by (k_0, k_1) and $k = k_0 + k_1$ respectively, usually assuming that $k_1 \leq k_0$ holds. We may sometimes choose to represent a necklace S of type (k_0, k_1) by fixing an unfolding $\tilde{S}$ of S and writing it out explicitly as the string $10^{j_1}10^{j_2}\ldots 10^{j_{k_1}}$, where $j_1, j_2, \ldots j_{k_1}$ are nonnegative integers with $\sum_{i=1}^{k_1} j_i = k_0$. Figure 1 contains examples of binary necklaces of various types.

We let $N(k_0, k_1)$ denote the set of all binary necklaces of type (k_0, k_1). A natural first question of study is that of the size of $N(k_0, k_1)$. Indeed this has been studied, and the results of [1] give an explicit formula for this cardinality:

Lemma 2.2. *Let k_0, k_1 be any nonnegative integers. Then:*

$$\#N(k_0, k_1) = \frac{1}{k} \sum_{d | k_0, k_1} \phi(d) \binom{k/d}{k_1/d}$$

Of particular interest is the following special case:

Corollary 2.3. *For n any nonnegative integer, we get:*

$$\#N(n+1,n) = \frac{1}{n+1}\binom{2n}{n}$$

We recognize this value as the nth Catalan number and are hence motivated to find a bijection between these binary necklaces and some suitable Catalan object. An answer exists in the simple but powerful *cycle lemma* of [2]: For every binary necklace S of type $(n+1,n)$, there is a unique $r \in \mathbb{Z}/k\mathbb{Z}$ such that $r \cdot S$ is a "modified Dyck word" of length $2n+1$: a sequence on the alphabet $\{0,1\}$ that has one more 0 than 1, ends with a 0, and such that any initial segment has no more 1's than 0's.

We are interested in studying a certain subset of $N(k_0,k_1)$ whose elements satsify an "equidistribution" property, say, in the sense of minimzing the *discrepancy* defined by Erdős and Turán [4]. Specifically, we ask: Which necklaces $S \in N(k_0,k_1)$ have the property that the distribution of 1's and 0's in any arc of S approximates the distribution of 1's and 0's in the entirety of S? This definition can be seen as a strenghtening of the notion of *balance* given in [7].

Definition 2.4. A binary necklace S of type (k_0,k_1) is called a *equidistributed* if, for any unfolding $\tilde{S}$ of S and any indices $m,n \in \mathbb{Z}/k\mathbb{Z}$, we get

$$(*) \qquad \left|\#(\tilde{S}^{-1}(1)\cap[m,n)) - \frac{k_1}{k}d(m,n)\right| < 1$$

Since $\#(\tilde{S}^{-1}(1)\cap[m,n))$ is an integer, this condition is equivalent to requiring that for any unfolding $\tilde{S}$ of S and any indices $m,n \in \mathbb{Z}/k\mathbb{Z}$, we get either

$$\#(\tilde{S}^{-1}(1)\cap[m,n)) = \left\lfloor \frac{k_1}{k}d(m,n)\right\rfloor \text{ or } \#(\tilde{S}^{-1}(1)\cap[m,n)) = \left\lceil \frac{k_1}{k}d(m,n)\right\rceil.$$

A binary necklace $S \in N(k_0,k_1)$ fails to be equidistributed when there exists some interval containing the "wrong" number of 0's or the "wrong" number of 1's; taking the complement of these intervals, we see that any such S must have some interval containing "too many" 1's, "too few" 1's, "too many" 0's, and "too few" 0's. In this way, non-equidistributed necklaces will provide a great deal of information to be used constructively. Moreover, this implies that a binary necklace is equidistributed if and only if its bitwise complement is equidstributed.

First we consider existence of equidistributed necklaces of a given type (k_0,k_1). Clear examples come to mind when k_1 divides k_0, one of which is given in Figure 1 (left). We make this observation precise in the following result:

Lemma 2.5. *Suppose that k_1 divides k_0. Then the necklace $R \in N(k_0,k_1)$ corresponding to the following unfolding is equidistributed:*

$$\tilde{R} = \left(10^{\frac{k_0}{k_1}}\right)^{k_1}$$

Proof. Let $\tilde{R}$ be the given unfolding and take any indices $m,n \in \mathbb{Z}/k\mathbb{Z}$. Define $a = \#(\tilde{R}^{-1}(1)\cap[m,n))$ and let us write $\tilde{R}[m,n) = 0^{j_0}10^{j_1}\dots 10^{j_a}$ so that we get $j_i = \frac{k_0}{k_1}$ for all $i \in \{1,\dots a-1\}$ and $0 \le j_i \le \frac{k_0}{k_1}$ for $i = 0,a$. Then we may bound:

$$\left|a - \frac{k_1}{k}d(m,n)\right| = \left|a - \frac{k_1}{k}(a + \sum_{i=0}^{a} j_i)\right| = \frac{1}{k}\left|k_0 - k_1(j_0 + j_a)\right| \tag{1}$$

But note that that $0 \leq j_0, j_a \leq \frac{k_0}{k_1}$ implies $|k_0 - k_1(j_0 + j_a)| \leq k_0$, so we can further bound (1) by:

$$\left|a - \frac{k_1}{k}d(m,n)\right| = \frac{k_0}{k} < 1 \tag{2}$$

Therefore, R is equidistributed. □

When k_1 does not divide k_0, the existence of equidistributed necklaces of type (k_0, k_1) is not quite so obvious. Nonetheless, we are able to provide a construction that holds in general. It should be noted that, when k_0 and k_1 are coprime, this process constructs a word that is cyclically equivalent to the unique Christoffel word of type (k_0, k_1) (see Exercise 1.7 of [6]). For illustrations of this process and of the result of this process, see Figure 2 and Figure 1 (right), respectively.

Proposition 2.6. *For any type (k_0, k_1) and any irrational number $\gamma > 0$, define the binary sequence:*

$$\tilde{R}[i] = \begin{cases} 0 & \text{if } \lfloor \frac{ik_0}{k} \rfloor \frac{1}{k_0} + \gamma > \lfloor \frac{ik_1}{k} \rfloor \frac{1}{k_1} \\ 1 & \text{otherwise} \end{cases}$$

Then the binary necklace R corresponding to $\tilde{R}$ is equidistributed.

Proof. Consider the path $(x(t), y(t)) = (k_0 t, k_1 t + \gamma)$ for $t \in [0, 1)$. Observe that $\tilde{R}$ is exactly the binary sequence which records a 0 whenever $x(t)$ is integer-valued and records a 1 whenever $y(t)$ is integer-valued. Since γ is irrational, these cases never coincide. That is, there is a map $i \mapsto t_i$ for which $\tilde{R}[i]$ agrees with whichever of $x(t_i)$ or $y(t_i))$ is integer-valued.

Now assume for the sake of contradiction that R is not equidistributed; that is, there exist indices $m, n \in \mathbb{Z}/k\mathbb{Z}$ such that $a = \#(\tilde{R}^{-1}(1) \cap [m, n))$ satisfies

$$a > \frac{k_1}{k}d(m,n) + 1 \qquad \text{and} \qquad d(m,n) - a < \frac{k_1}{k}d(m,n) - 1.$$

Consider the line segment from $(x(t_m), y(t_m))$ to $(x(t_n), y(t_n))$. Note that we have $y(t_n) - y(t_m) \geq a - 1 > \frac{k_1}{k}d(m,n)$ and $x(t_n) - x(t_m) < d(m,n) - a + 1 < \frac{k_1}{k}d(m,n)$. This implies that the segment has slope

$$\frac{y(t_n) - y(t_m)}{x(t_n) - x(t_m)} > 1. \tag{3}$$

But also consider the complement of this interval, $\tilde{R}[n, m)$. It must follow that $b = \#(\tilde{R}^{-1}(1) \cap [n, m)) = k_1 - a$ satisfies

$$b < \frac{k_1}{k}d(n,m) - 1 \qquad \text{and} \qquad d(n,m) - b > \frac{k_1}{k}d(n,m) + 1.$$

This interval corresponds to the line segment from $(x(t_n), y(t_n))$ to $(x(t_m + 1), y(t_m + 1))$. Hence, the bounds $y(t_m + 1) - y(t_n) \leq b + 1 < \frac{k_1}{k}(n, m)$ and $x(t_m + 1) - x(t_n) \geq d(n, m) - b - 1 > \frac{k_1}{k} d(n, m)$ give that the line segment has slope

$$\frac{y(t_m + 1) - y(t_n)}{x(t_m + 1) - x(t_n)} < 1. \tag{4}$$

Since these segments are contained within the same line, they must have the same slope. However, this is clearly impossible by (3) and (4), so R must be equidistributed. □

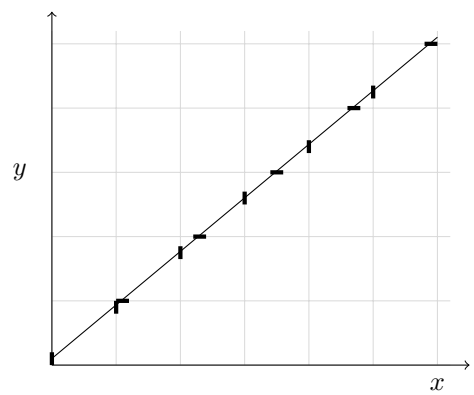

FIGURE 2. The construction given in Proposition 2.6 for γ small and $(k_0, k_1) = (6, 5)$. The resulting binary necklace corresponds to the string 00101010101.

The next few definitions and results will enrich our theory of binary necklaces and eventually lead to the desired existence and uniqueness result.

Definition 2.7. We say that a necklace $S \in N(k_0, k_1)$ has *balanced gaps* if any unfolding $\tilde{S} = 10^{j_1}10^{j_2} \ldots 10^{j_{k_1}}$ has, for all i, either $j_i = \lfloor \frac{k_0}{k_1} \rfloor$ or $\lceil \frac{k_0}{k_1} \rceil$.

Note that if $S \in N(k_0, k_1)$ has balanced gaps, then we necessarily have that the number of i with $j_i = \lfloor \frac{k_0}{k_1} \rfloor$ is $k_1 - (k_0 \mod k_1)$ and the number of i with $j_i = \lceil \frac{k_0}{k_1} \rceil$ is $k_0 \mod k_1$. We use $BGN(k_0, k_1)$ to denote the subset of $N(k_0, k_1)$ whose elements have balanced gaps. The following properties of balanced gap necklaces are quite natural:

Lemma 2.8. *If k_1 divides k_0, then $BGN(k_0, k_1)$ consists of one necklace, and it is equidistributed.*

Proof. If k_1 divides k_0, then $\lfloor \frac{k_0}{k_1} \rfloor = \lceil \frac{k_0}{k_1} \rceil = \frac{k_0}{k_1}$. Hence, the only possible necklace in $BGN(k_0, k_1)$ is the S corresponding to the construction given in Proposition 2.5, which we know is equidistributed. □

Lemma 2.9. *If $S \in N(k_0, k_1)$ is equidistributed, then it has balanced gaps.*

Proof. Let $\tilde{S}$ be any unfolding of $S \in N(k_0, k_1)$ and consider any substring of the form 10^{j_i} in $\tilde{S}$. Use $(*)$ on this substring to get the following cases: either $\lfloor \frac{k_1}{k}(j_i+1) \rfloor = 1$ or $\lceil \frac{k_1}{k}(j_i+1) \rceil = 1$

The first case is equivalent to $1 \leq \frac{k_1}{k}(j_i+1) < 2$. In particular, this gives $\frac{k_0}{k_1} + 1 = \frac{k}{k_1} \leq j_i + 1$, which implies $\frac{k_0}{k_1} \leq j_i$. Then use $(*)$ on the substring 0^{j_i} to get $\lfloor \frac{k_1}{k} j_i \rfloor = 0$. In particular, this gives $\frac{k_1}{k} j_i < 1$, which implies $j_i < \frac{k}{k_1} = \frac{k_0}{k_1} + 1$. Combine these to get $\frac{k_0}{k_1} \leq j_i < \frac{k_0}{k_1} + 1$, which is exactly $j_i = \lfloor \frac{k_0}{k_1} \rfloor$.

The second case is equivalent to $0 < \frac{k_1}{k}(j_i+1) \leq 1$. In particular, this gives $j_i + 1 \leq \frac{k}{k_1} = \frac{k_0}{k_1} + 1$, which implies $j_i \leq \frac{k_0}{k_1}$. Then use $(*)$ on the substring $10^{j_i}1$ to get $\lfloor \frac{k_1}{k}(j_i+2) \rfloor = 2$ or $\lceil \frac{k_1}{k}(j_i+2) \rceil = 2$. In either case, this gives $1 < \frac{k_1}{k}(j_i+2)$, which implies $\frac{k_0}{k_1} - 1 < j_i$. Combine these to get $\frac{k_0}{k_1} - 1 < j_i \leq \frac{k_0}{k_1}$, which is exactly $j_i = \lceil \frac{k_0}{k_1} \rceil$. □

All this discussion of balanced gap necklaces leads us to the following construction, which should be seen as the key step in the development of this theory.

Definition 2.10. For any necklace $S \in BGN(k_0, k_1)$ with $k_1 > 0$ and k_1 not dividing k_0, construct a new binary necklace by substituting

$$10^{\lfloor \frac{k_0}{k_1} \rfloor} \mapsto 0 \text{ and } 10^{\lceil \frac{k_0}{k_1} \rceil} \mapsto 1. \tag{5}$$

The resulting necklace is called the *compression* of S, denoted compress(S). If it is the case that k_1 divides k_0, then we use the convention that compress(S) is the binary necklace of length k_1 which consists only of 0's.

Lemma 2.11. *For $k_1 > 0$, the map*

$$\textit{compress} : BGN(k_0, k_1) \to N(k_1 - (k_0 \mod k_1), k_0 \mod k_1)$$

is a bijection.

Proof. Consider the map sending

$$0 \mapsto 10^{\lfloor \frac{k_0}{k_1} \rfloor} \text{ and } 1 \mapsto 10^{\lceil \frac{k_0}{k_1} \rceil}. \tag{6}$$

This is an explicit construction of the inverse of (5) on $BGN(k_0, k_1)$. □

Remark 2.12. The type of a compressed necklace depends only on the type of the original necklace, not on the particular necklace compressed. Moreover, the number of 1's in the compressed necklace is strictly less than the number of 1's in the original necklace, since $k_0 \mod k_1 < k_1$.

It should be noted that, to be fully precise with this definition, each necklace type should have its own compress map (and, more importantly, its own inverse). This abuse of notation will be convenient for our mathematical purposes, but it will be addressed more carefully in Section 3. Now we can proceed to the main lemma of this theory:

Lemma 2.13. *A necklace $S \in BGN(k_0, k_1)$ is equidistributed if and only if compress(S) is equidistributed.*

Proof. The claim is easy when k_1 divides k_0: If we have $S \in BGN(k_0, k_1)$ with k_1 dividing k_0, then compress(S) has type $(k_1, 0)$ which is equidistributed. Conversely, if we have compress$(S) \in N(k_1, 0)$ with k_1 dividing k_0, then S is equidistributed by Proposition 2.8. Now we proceed assuming that k_1 does not divide k_0, which allow us to use the useful identity:

$$\left\lceil \frac{k_0}{k_1} \right\rceil - \left\lfloor \frac{k_0}{k_1} \right\rfloor = 1 \tag{7}$$

First suppose that $S \in BGN(k_0, k_1)$ is not equidistributed. Then there exists some unfolding $\tilde{S}$ of S and some interval $[m, n)$ such that we can write $\tilde{S}[m, n) = 10^{j_1}1 \dots 10^{j_a}1$ with $j_i = \lfloor \frac{k_0}{k_1} \rfloor$ or $\lceil \frac{k_0}{k_1} \rceil$ for all i, and we get

$$a + 1 \geq \frac{k_1}{k}\left(a + 1 + \sum_{i=1}^{a} j_i\right) + 1. \tag{8}$$

Letting r denote the number of $i \in \{1, \dots a\}$ for which $j_i = \lceil \frac{k_0}{k_1} \rceil$, we see that (8) can be expressed as

$$\begin{aligned} a + 1 &\geq \frac{k_1}{k}\left(a + 1 + (a - r)\left\lfloor \frac{k_0}{k_1} \right\rfloor + r\left\lceil \frac{k_0}{k_1} \right\rceil\right) + 1 \\ &= \frac{k_1}{k}\left(a + 1 + a\left\lfloor \frac{k_0}{k_1} \right\rfloor + r\right) + 1, \end{aligned} \tag{9}$$

where the equality follows from identity (7). Then check that (9) rearranges to

$$r \leq \left(\frac{k_0}{k_1} - \left\lfloor \frac{k_0}{k_1} \right\rfloor\right)a - 1 = \left(\frac{k_0 \mod k_1}{k_1}\right)a - 1. \tag{10}$$

But observe that, by definition, r is exactly the number of 1's in the image of $\tilde{S}[m, n)$ under the compress map. Therefore, compress(S) is not equidistributed.

For the second direction, suppose that $S \in BGN(k_0, k_1)$ is such that $C =$ compress(S) is not equidistributed. Then there must exist an unfolding $\tilde{C}$ of C and an interval $[m, n)$ such that, for $r = \#(\tilde{C}^{-1}(1) \cap [m, n))$ and $a = n - m$, we get

$$r \leq \frac{k_0 \mod k_1}{k_1}a - 1 = \left(\frac{k_0}{k_1} - \left\lfloor \frac{k_0}{k_1} \right\rfloor\right)a - 1. \tag{11}$$

But observe that (11) is equivalent to (10), so it rearranges to (9). That is,

$$\begin{aligned} a + 1 &\geq \frac{k_1}{k}\left(a + 1 + a\left\lfloor \frac{k_0}{k_1} \right\rfloor + r\right) + 1 \\ &= \frac{k_1}{k}\left(a + 1 + (a - r)\left\lfloor \frac{k_0}{k_1} \right\rfloor + r\left\lceil \frac{k_0}{k_1} \right\rceil\right) + 1, \end{aligned} \tag{12}$$

where, again, we used identity (7) in the equality. Then note that there exists an unfolding $\tilde{S}$ of S and an interval $[m', n')$ such that $\tilde{C}[m, n)$ is the image of $\tilde{S}[m', n') = 10^{j_1} \dots 10^{j_a}$ under the compress map, which necessarily implies $a + (a - r)\lfloor \frac{k_0}{k_1} \rfloor + r\lceil \frac{k_0}{k_1} \rceil = n' - m'$. Then (12) is equivalent to

$$a+1 \geq \frac{k_1}{k}((n'-m')+1)+1. \tag{13}$$

Let us consider this inequality with respect to the string $\tilde{S}[m', n'+1) = 10^{j_1} \dots 10^{j_a}1$. On the left side, $a+1$ is equal to $\#(\tilde{S}^{-1}(1) \cap [m', n'+1))$, and, on the right side $(n'-m')+1$, is equal to the length of the interval in question. That is, (13) implies that S is not equidistributed, which completes the proof. □

Theorem 2.14. *A necklace $S \in N(k_0, k_1)$ is equidistributed if and only if we get compress$^r(S) \in N(k_0', 0)$ for some nonnegative integer r and positive integer k_0'.*

Proof. First, let $S \in N(k_0, k_1)$ be an equidistributed necklace. Then use Proposition 2.13 inductively to see that every necklace that is an element of the sequence $\{\text{compress}^0(S), \text{compress}^1(S), \text{compress}^2(S), \dots\}$ is equidistributed. Denoting the respective types of these necklaces by $\{(k_0^0, k_1^0), (k_0^1, k_1^1), ((k_0^2, k_1^2), \dots\}$, Remark 2.12 guarantees that we must have $k_1^0 > k_1^1 > k_1^2 > \dots$. As this is a strictly decreasing sequence of positive integers, there must be some nonnegative integer r such that $k_1^r = 0$ holds. Then we have compress$^r(S) \in N(k_0^r, 0)$, as desired.

Conversely, suppose that $S \in N(k_0, k_1)$ is such that compress$^r(S) = N(k_0', 0)$ holds for some nonnegative integer r and some positive integer k_0'. Use Proposition 2.13 inductively to see that $\{\text{compress}^r(S), \text{compress}^{r-1}(S), \dots \text{compress}^0(S)\}$ is a finite sequence of equidistributed necklaces. In particular, we see that the element compress$^0(S) = S$ is equidistributed, as desired. □

By Remark 2.12, the integer r above depends only on the type (k_0, k_1) and not on the specific necklace S. This observation leads us to the promised uniqueness result:

Theorem 2.15. *For every type (k_0, k_1), there exists a unique equidistributed necklace in $N(k_0, k_1)$.*

Proof. Suppose that $S_1, S_2 \in N(k_0, k_1)$ were both equidistributed. Theorem 2.14 guarantees a nonnegative integer r such that compress$^r(S_1) =$ compress$^r(S_2)$ holds. But by Lemma 2.11, every compress map is injective, so the composition of a finite number of them is also injective. Therefore, $S_1 = S_2$. □

Having already proven an existence result in Proposition 2.6, we have found all the possible equidistributed necklaces!

3. Algorithm

Finally, we turn to the task of computationally checking whether a given binary necklace is equidistributed. Theorem 2.14 suggests a recursive algorithm for doing this. While it is slow in its current form, its runtime can be improved dramatically by making the following modification: If necessary, flip the bits of compress(S) in order to maintain that the precondition $k_1 \leq k_0$ holds at every function call. In particular, types will be mapped under compress as follows: If $k_0 \mod k_1 \geq \frac{k_1}{2}$, then $(k_0, k_1) \mapsto (k_0 \mod k_1, k_1 - (k_0 \mod k_1))$; if $k_0 \mod k_1 < \frac{k_1}{2}$, then $(k_0, k_1) \mapsto (k_1 - (k_0 \mod k_1), k_0 \mod k_1)$. We record this procedure precisely in Algorithm 1.

Algorithm 1 Checking for equidistribution of a binary necklace recursively

```
1: procedure IsEqui(S)
2:     k ← length(S)
3:     k_1 ← sum(S)
4:     k_0 ← k − k_1
5:     return IsEquiHelper(S, k_0, k_1, k)
6:
7: procedure IsEquiHelper(S, k_0, k_1, k)
8:     if S ∉ BGN(k_0, k_1) then
9:         return false
10:    else
11:        if k_1 = 0 or k_1 divides k_0 then
12:            return true
13:        S' ← compress(S, k_0, k_1)
14:        if k_0 mod k_1 < k_1/2 then
15:            return IsEquiHelper(S', k_1 − (k_0 mod k_1), k_0 mod k_1, k_1)
16:        else
17:            S' ← S' ⊕ 1^{k_1}                                  ▷ Flip all of the bits
18:            return IsEquiHelper(S', k_0 mod k_1, k_1 − (k_0 mod k_1), k_1)
```

Theorem 3.1. *For a necklace $S \in N(k_0, k_1)$, Algorithm 1 returns whether or not S is equidistributed. It runs in $\Theta(k)$ time and in $O(k_1)$ space. The depth of its function call stack is $O(\log_2 k_1)$.*

Proof. Theorem 2.14 proves correctness of the unmodified algorithm. To see that the modification does not alter the correctness, simply recall that $S \in N(k_0, k_1)$ is equidistributed if and only if $S \oplus 1^k$ is equidistributed.

To see the time complexity, observe that lines 8 and 13 both run in $O(k)$ time, and that all other lines run in $O(1)$ time. Since the modification guarantees that $k_0 + k_1$ drops by a factor of at least 2 at every function call, we see that the total time complexity is $O(k)+O(\frac{k}{2})+O(\frac{k}{4})+\cdots = O(k)$. Since line 2 runs in $\Omega(k)$ time, we get a total time complexity of $\Theta(k)$. Likewise, lines 8 and 13 run in $O(k_1)$ space and all other lines are $O(1)$ space. Since the modification guarantees that k_1 drops by a factor of at least 2 at every function call, we see that the total space complexity is $O(k_1)+O(\frac{k_1}{2})+O(\frac{k_1}{4})+\cdots = O(k_1)$. The modification also guarantees that the function call stack has a depth of $O(\log_2 k_1)$. □

4. Acknowledgements

This paper began during my residence at the Institute for Computational and Experimental Research in Mathematics (ICERM) in the Summer of 2017, and I am very grateful to the students and faculty who helped me formulate this problem. Specific thanks are due to Henry Adams, Samir Chowdhury, and Peter Eastwood. I also owe thanks to Atticus Geiger, Prasanna Ramakrishnan, and Bora Uyumazturk at Stanford for many helpful conversations.

References

[1] M. Jibladze A. Elashvili and D. Pataraia. Combinatorics of necklaces and "hermite reciprocity". *Journal of Algebraic Combinatorics*, 10(2):273–188, 1999.

[2] N. Derchowitz and S. Zak. The cycle lemma and some applications. *European Journal of Combinatorics*, 11(1):35–40, 1990.
[3] P. Erdős and R. R. Hall. On the angular distribution of gaussian integers with fixed norm. *Discrete Mathematics*, 200(1):87–94, 1999.
[4] P. Erdős and P. Turán. On a problem in the theory of uniform distribution I, II. *Koninklijke Nederlandse Akademie van Wetenschappen*, 51:1146–1154,1262–1269, 1948.
[5] J. R. Hollos and S. Hollos. *Creating Rhythms*. Abrazol Publishing, 2014.
[6] C. Reutenauer J. Berstel, A. Lauve and F. Saliola. Combinatorics on words: Christoffel words and repetitions in words, 2008.
[7] M. Jiang. On the sum of distances along a circle. *Discrete Mathematics*, 308(10):2038–2045, 2008.
[8] Godfried T. Toussaint. *The Geometry of Musical Rhythm: What Makes a Good Rhythm Good?* Chapman and Hall/CRC, 2013.
[9] H. Weyl. Über die Gleichverteilung von Zahlen modulo Eins. *Annals of Mathematics*, 77:313–352, 1916.

Department of Mathematics, Stanford University, Stanford, CA 94305, United States
E-mail address: aqjaffe@stanford.edu